Department of Chemical Engineering, Chulalongkorn University — teaching

SO3 CONVERTER VER 1

Operator Training Simulator · sulfur-burning H_2SO_4 plant, converter section

CAUSE AND EFFECT DIAGRAM

Safety instrumented system (I-1 … I-6) and alarms

Document No. SO3-V1-IN-CE-001 · Rev R0

Causes	38 rows: 6 trips (I-1 … I-6), 30 alarm limits, 2 sequence / interlock rows; 2 spare rows
Effects	10 columns: XV-103, FV-103, AIC-104, K-101, H-101, P-301B, SIS, 1ST, ALM, TRP
Final element	XV-103 sulfur shut-off valve (every SIS trip); the ESD also stops K-101 and H-101
Sheets	1 cover (A4) + 1 cause and effect sheet (A3 landscape)
Related documents	SO3-V1-IN-SRS-001 Safety requirements specification (SRS of SIF tab); LOPA tab; P&ID tab of the simulator (screen drawing); trip logic src/model.js tripConditions / bmsStep / bmsTrip
Basis	Trip settings, delays and effects as coded in the simulator (tripConditions, bmsStep, bmsTrip / sulfurOff); alarm limits from the ALM table. Checked in the C&E tab of the simulator.

Revision record

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKER	APPROVER
R0	19-Oct-26	Issued for teaching	Apinan Soottitantawat	—	—

Issued for teaching. Not for construction or operation of a real plant.

CAUSE AND EFFECT DIAGRAM — SIS AND ALARMS

SO3 Converter Ver 1 · converter section · SIS I-1 … I-6, alarm limits and sequence interlocks

SO3-V1-IN-CE-001

Rev R0 · Sheet 1 of 1

CAUSES								EFFECTS										
ITEM	TAG No.	TAG SERVICE DESCRIPTION	TRIP / ALARM DIRECTION	SETTING	ENG. UNIT	DELAY	NOTE	XV-103 Sulfur shut-off valve (SIS final element)	FV-103 Sulfur flow valve (FIC-103 MAN 0 %)	AIC-104 SO_2 controller to MAN	K-101 Main air blower	H-101 Preheat burner (start-up only)	P-301B Spare IPA acid pump (HS-301B armed)	SIS SIS state (latched until reset)	1ST First-out annunciation	ALM DCS alarm and horn	TRP SIS trip annunciation and horn
							TRIP ACTION →	C	C	MAN	STOP	OFF	START	TP	F	A	A
001	I-6 ESD	Emergency shutdown push-button (SIS screen)	Pushed	—	—	0 s	3	X	X	X	X	X		TP	X		X
002	FI-100	Total air from K-101 (FT-100) low alarm	Low	200	kNm³/h		1									X	
003	FI-100	Total air from K-101 (FT-100) high alarm	High	265	kNm³/h		1									X	
004	FSLL-100	Total air low-low trip, I-1 (FT-100)	Low-Low	90	kNm³/h	3 s	2, 6	X	X	X				TP	X		X
005	FI-100	H-101 preheat burner low-air interlock (burner protection, not SIS)	Low	60	kNm³/h	0 s	4					X				X	
006	AI-105	O_2 into bed 1 low alarm	Low	8	% O_2		1									X	
007	FIC-103	Molten sulfur to F-101 low alarm	Low	35	t/h		1									X	
008	FIC-103	Molten sulfur to F-101 high alarm	High	44	t/h		1									X	
009	AIC-104	SO_2 into bed 1 (AI-104) low alarm	Low	11	% SO_2		1									X	
010	AIC-104	SO_2 into bed 1 (AI-104) high alarm	High	12.8	% SO_2		1									X	
011	TI-107	WHB B-101 gas outlet high alarm	High	340	°C		1									X	
012	PI-200	Converter inlet pressure high alarm	High	42	kPa g											X	
013	TIC-201	Bed 1 inlet temperature (TT-201) low alarm	Low	405	°C		1									X	
014	TIC-201	Bed 1 inlet temperature (TT-201) high alarm	High	435	°C		1									X	
015	TI-203	Bed 1 exit temperature (TE-203) high alarm	High	640	°C		7									X	
016	TSHH-203	Bed 1 exit high-high trip, I-2 (TE-203)	High-High	660	°C	5 s	7, 8	X	X	X				TP	X		X
017	PDI-204	Bed 1 pressure drop high alarm	High	8	kPa											X	
018	TIC-211	Bed 2 inlet temperature low alarm	Low	430	°C		1									X	
019	TIC-211	Bed 2 inlet temperature high alarm	High	460	°C		1									X	
020	TIC-221	Bed 3 inlet temperature low alarm	Low	425	°C		1									X	
021	TIC-221	Bed 3 inlet temperature high alarm	High	455	°C		1									X	
022	TIC-231	Bed 4 inlet temperature low alarm	Low	405	°C		1									X	
023	TIC-231	Bed 4 inlet temperature high alarm	High	445	°C		1									X	
024	TIC-241	Bed 5 inlet temperature low alarm	Low	410	°C		1									X	
025	TIC-241	Bed 5 inlet temperature high alarm	High	445	°C		1									X	
026	TIC-251	Gas to IPA T-301 low alarm	Low	165	°C		1									X	
027	TIC-251	Gas to IPA T-301 high alarm	High	230	°C		1									X	
028	FI-253	Economizer BFW flow low alarm	Low	110	t/h		1									X	
029	FI-301	IPA acid circulation (FT-301) low alarm	Low	1200	m³/h		1									X	
030	FI-301	P-301B auto-start on low IPA acid flow (HS-301B armed; one shot)	Low	960	m³/h	3 s	5						X			X	
031	FSLL-301	IPA acid circulation low-low trip, I-3 (FT-301)	Low-Low	800	m³/h	10 s	2, 5	X	X	X				TP	X		X
032	AI-303	IPA acid strength low alarm	Low	98	% H_2SO_4											X	
033	TI-304	IPA acid temperature high alarm	High	95	°C											X	
034	FI-302	FAT acid circulation (FT-302) low alarm	Low	750	m³/h		1									X	
035	FSLL-302	FAT acid circulation low-low trip, I-4 (FT-302)	Low-Low	500	m³/h	10 s	2	X	X	X				TP	X		X
036	AIT-306	Stack SO_2 (permit 500 ppm) high alarm	High	500	ppm		1									X	
037	ASHH-306	Stack SO_2 high-high trip, I-5 (AIT-306)	High-High	3000	ppm	300 s	2, 9	X	X	X				TP	X		X
038	AIT-307	Stack acid mist high alarm	High	40	mg/Nm³		1									X	
039	SPARE																
040	SPARE																

Issued for teaching (Operator Training Simulator). Settings checked against the simulator code; not for a real plant.

NOTES

1. Alarm suppressed while sulfur is off (SIS not in RUN), so a stopped plant does not stand in alarm.
2. Armed only while XV-103 is open (sulfur admitted); FSLL-100/301/302 and ASHH-306 do not block the SIS reset.
3. Manual trip, no delay. Same effects as any SIS trip, and it also stops K-101 and H-101 (bmsTrip with esd).
4. Local burner interlock in the model step, not an SIS function: H-101 is shut off when total air falls below 60 kNm³/h or K-101 stops.
5. FT-301 serves FI-301, the P-301B auto-start (below 60 % of 1600 m³/h for 3 s) and FSLL-301 (below 50 %). The auto-start disarms itself after one attempt.
6. A low total-air trip is reached only after the furnace is already sulfur-rich at full rate (stoichiometric air ≈ 137 kNm³/h combustion air): see SRS SIF-01 and LOPA L1.
7. TE-203 serves the TI-203 indication, its 640 °C alarm and TSHH-203: one failed element defeats both.
8. Always armed; a TSHH-203 cause still present blocks the SIS reset.
9. Inhibited for 60 min after every sulfur admission (start-up), then 5 min delay.
10. Action codes: C = close; MAN = controller forced to manual; STOP / OFF = stop the drive or burner; START = start the spare; TP = SIS tripped (latched; manual reset, then the start permissives); F = first-out; A = activate.
11. Time delays act in the logic solver (bmsStep); trip effects latch until reset. Air keeps flowing after a sulfur trip (except ESD and I-1), which sweeps the furnace and cools the catalyst.

LEGEND

- Trip cause (SIS, I-1 … I-6): tag in red, row shaded
- Sequence / interlock cause (not an SIS trip)
- Alarm cause (DCS alarm and horn only)

X = the cause drives the effect; the code in the trip action row says what the effect does. TP = SIS tripped (latched).

CLIENT	Department of Chemical Engineering, Chulalongkorn University — teaching
PROJECT	SO3 Converter Ver 1 · Operator Training Simulator (sulfur-burning H_2SO_4 plant, converter section)
DOCUMENT No.	**SO3-V1-IN-CE-001**
TITLE	**Cause and Effect Diagram — SIS and alarms**
P&ID	P&ID tab of the simulator (screen drawing); trip logic src/model.js tripConditions / bmsStep / bmsTrip

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